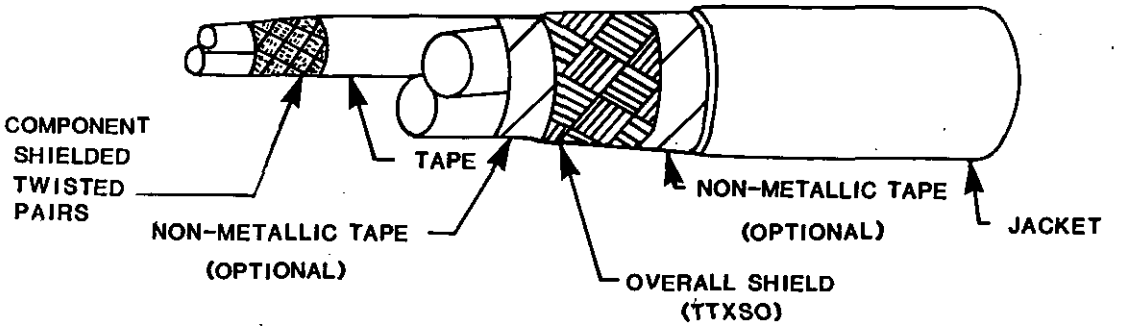
MIL-C-0024640/5A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/5
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR AWG 20, TYPE TTXS (INCLUDING VARIATION TYPES TTXSA AND TTXSO)

This military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24640. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24640.



SH 12752

COMPONENT SHIELDED PAIR: One black MIL-W-81044/12-20 wire twisted with one white, MIL-W-81044/12-20 wire except identification codes and methods shall be in accordance with MIL-C-24640. The shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (non-watertight).

(A) denotes changes.

FSC 6145

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(A) TABLE I. Construction details - unarmored.

Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
TTXS-2	2	0.038	0.335	0.361	74.3
TTXS-4	4	.041	.393	.424	115.0

(A) TABLE II. Construction details - armored.

Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
TTXSA-2	2	0.038	0.385	0.411	100.2
TTXSA-4	4	.041	.443	.474	150.4

(A) TABLE III. Construction details - overall shield.

Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
TTXSO-2	2	0.283	0.041	0.386	0.416	126.0
TTXSO-6	6	.418	.044	.506	.546	216.0
TTXSO-8	8	.510	.044	.587	.633	284.0
TTXSO-10	10	.553	.044	.627	.675	311.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

When this cable is manufactured with armor, the following additional requirements shall apply:

(a) Braided metal armor and paint shall be added.
(b) Cable surface marking will not be required.

When this cable is manufactured with an overall shield, the following additional requirements shall apply:

(a) An overall tin-coated copper braid shall be added.
(b) Braid coverage shall be not less than 85 percent determined in accordance with MIL-C-27500.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	10.07
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Conductor to shield	1000
Shield to shield (TTXSO only)	500
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity (TTXSO only)	No failure
Jacket flaws (TTXSO only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	50
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	45 ± 5
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	3.5
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor (type TTXSA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	--
Surface transfer impedance (type TTXSO only)	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	--
Decay time, maximum	--

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-N322-05)

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